

Lesson 4 Lesson-End Project

Designing Role Based Access Control

Project agenda: To implement role-based access control to define various roles

Description: You are working as the head of cloud security in an organization. You have been given a project to design authorization as part of cloud resource compliance and security. You have been asked to restrict access to different scopes depending on the employee's designation and role and to ensure you follow the principle of least privilege.

Tools required: Azure account with administrator access

Prerequisites: None

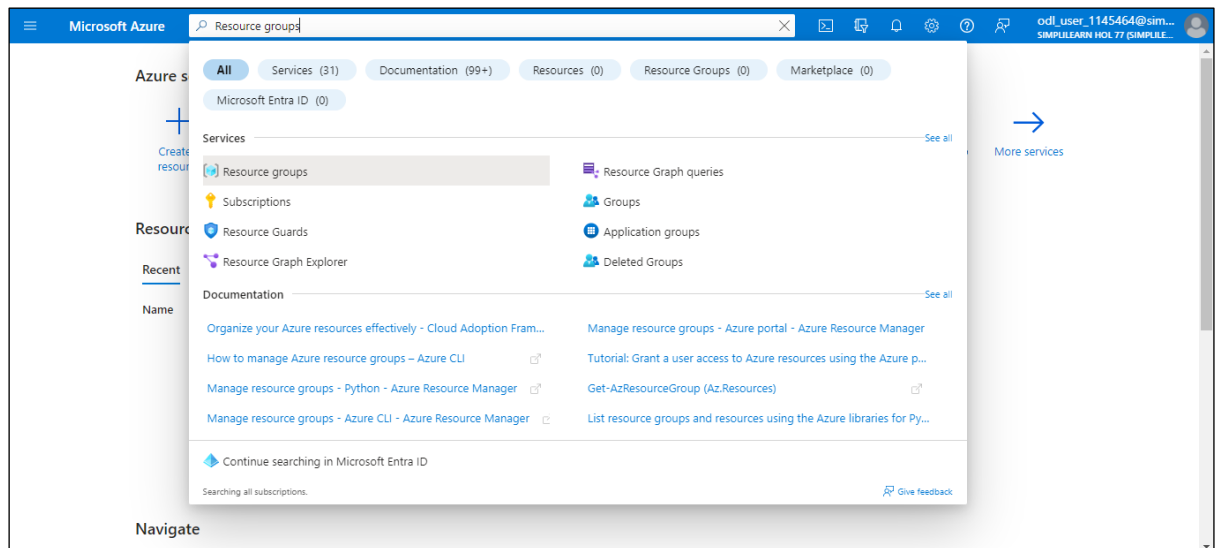
Expected deliverables: Add a screenshot showing the role-based access control applied on the resource group level to the groups of users having a similar designation

Steps to be followed:

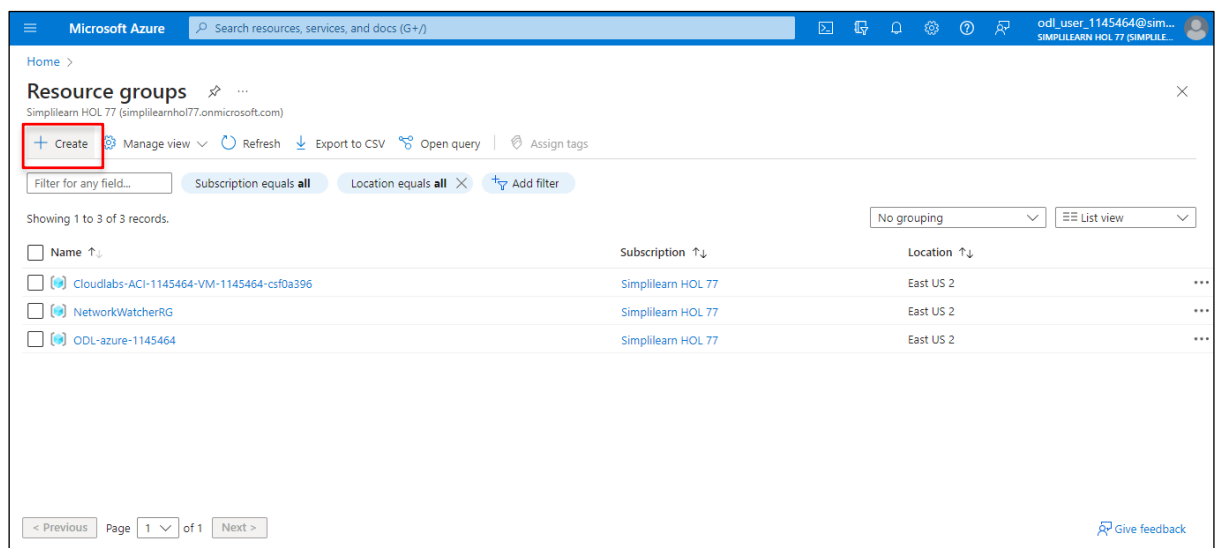
1. Create two resource groups in two different regions
2. Create two groups under two resource groups in Microsoft Entra ID
3. Create two users and add them to each of the resource groups in Microsoft Entra ID
4. Grant contributor and security assessment contributor access to the DEVELOPMENT and SECURITY ANALYST groups.

Step 1: Create two resource groups in two different regions

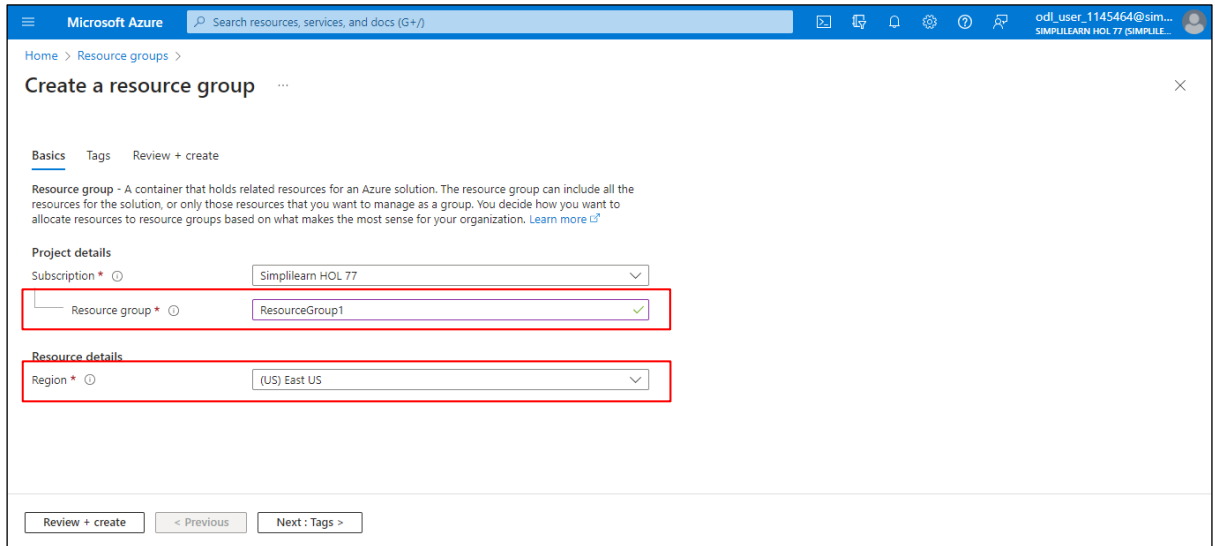
1.1 Log in to the Azure portal; search for and select Resource groups



1.2 In the Resource groups page, click on Create

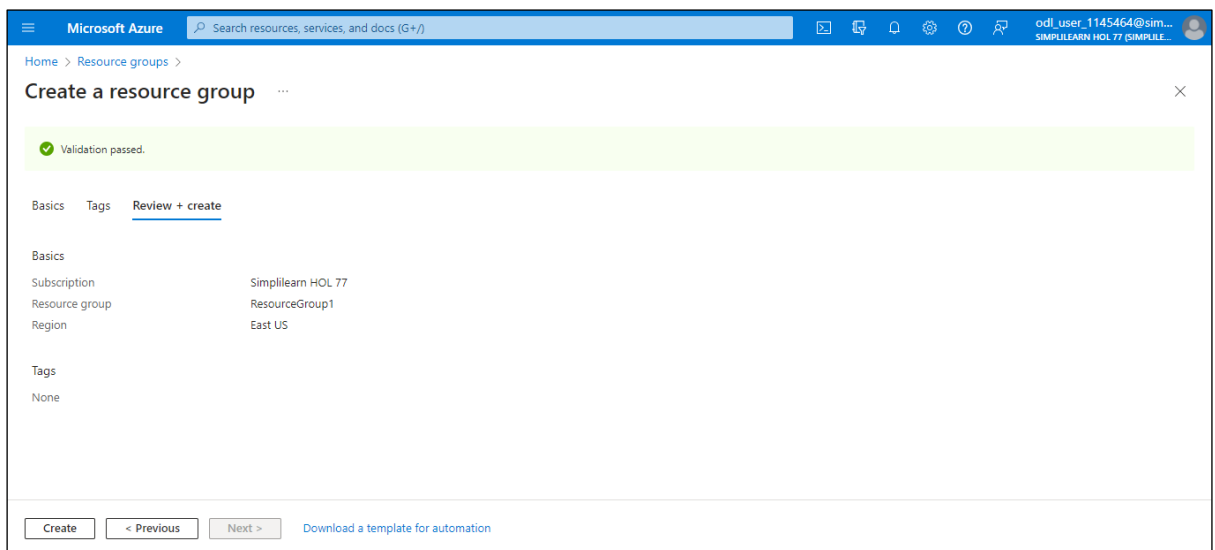


1.3 In the **Create a resource group** page, navigate to **Basics** tab. Under **Project details**, create a new **Resource group ResourceGroup1** and select the appropriate **Region (US) East US**. Then, click on **Review + create**.



The screenshot shows the 'Create a resource group' page in the Microsoft Azure portal. The 'Basics' tab is selected. Under 'Project details', the 'Subscription' is set to 'Simplilearn HOL 77'. The 'Resource group' is set to 'ResourceGroup1'. Under 'Resource details', the 'Region' is set to '(US) East US'. The 'Review + create' button is visible at the bottom.

1.4 In the **Review + create** tab, click on **Create** once the validation is passed



The screenshot shows the 'Create a resource group' page in the Microsoft Azure portal, now in the 'Review + create' tab. A green banner at the top indicates 'Validation passed.' The 'Basics' section shows the 'Subscription' as 'Simplilearn HOL 77', 'Resource group' as 'ResourceGroup1', and 'Region' as 'East US'. The 'Tags' section shows 'None'. The 'Create' button is visible at the bottom.

Now, the resource group is created.

Resource groups

Subscription equals **all** Location equals **all** Add filter

Showing 1 to 4 of 4 records.

Name	Subscription	Location
Cloudlabs-ACI-1145464-VM-1145464-csf0a396	Simplilearn HOL 77	East US 2
NetworkWatcherRG	Simplilearn HOL 77	East US 2
ODL-azure-1145464	Simplilearn HOL 77	East US 2
ResourceGroup1	Simplilearn HOL 77	East US

1.5 Similarly, create another **Resource group ResourceGroup2** in a different **Region (US)** **East US 2**

Create a resource group

Basics Tags Review + create

Project details

Subscription * Simplilearn HOL 77

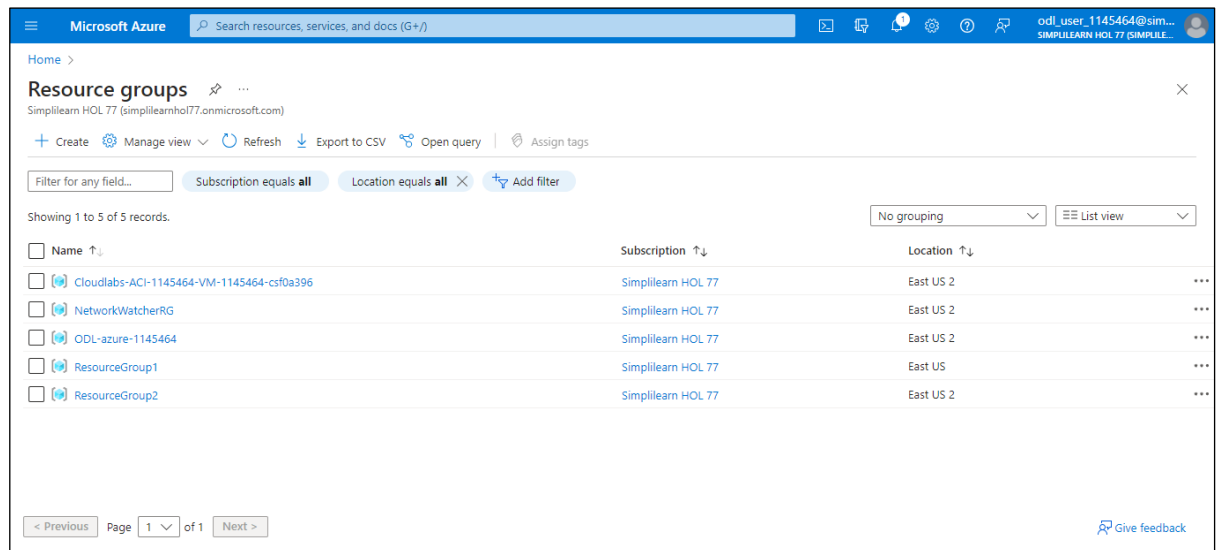
Resource group * ResourceGroup2

Resource details

Region * (US) East US 2

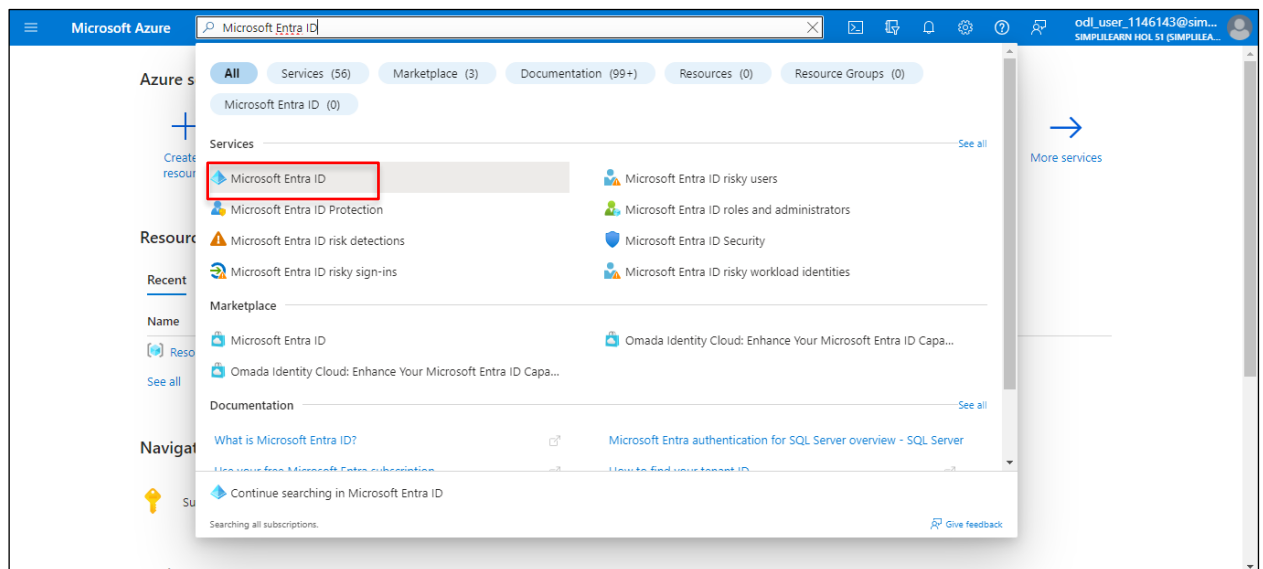
Review + create < Previous Next: Tags >

The two resource groups **ResourceGroup1** and **ResourceGroup2** are now created.

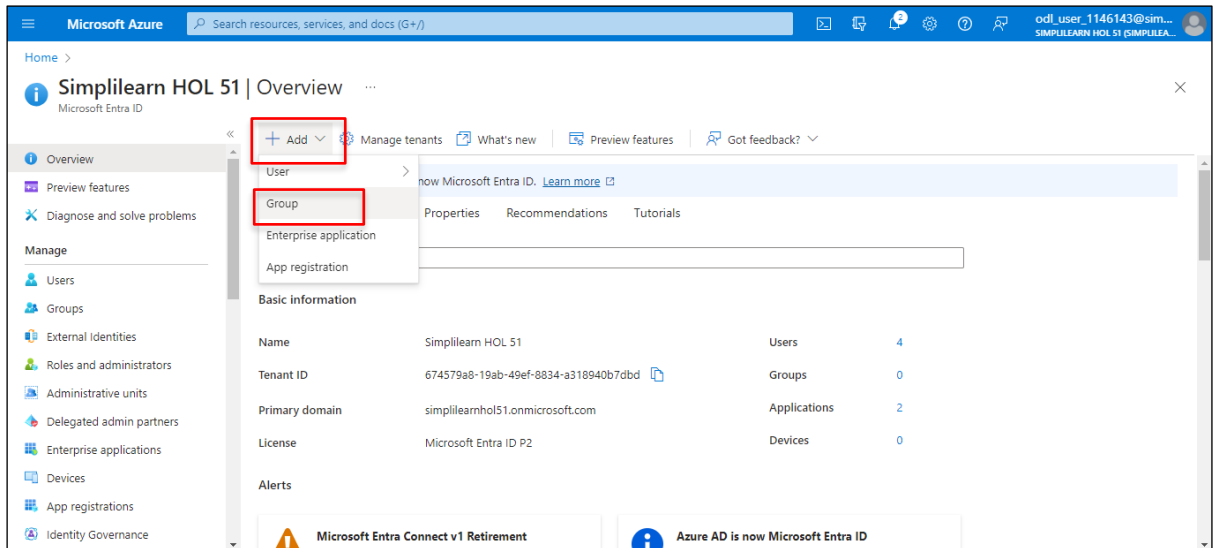


Step 2: Create two groups under two resource groups in Microsoft Entra ID

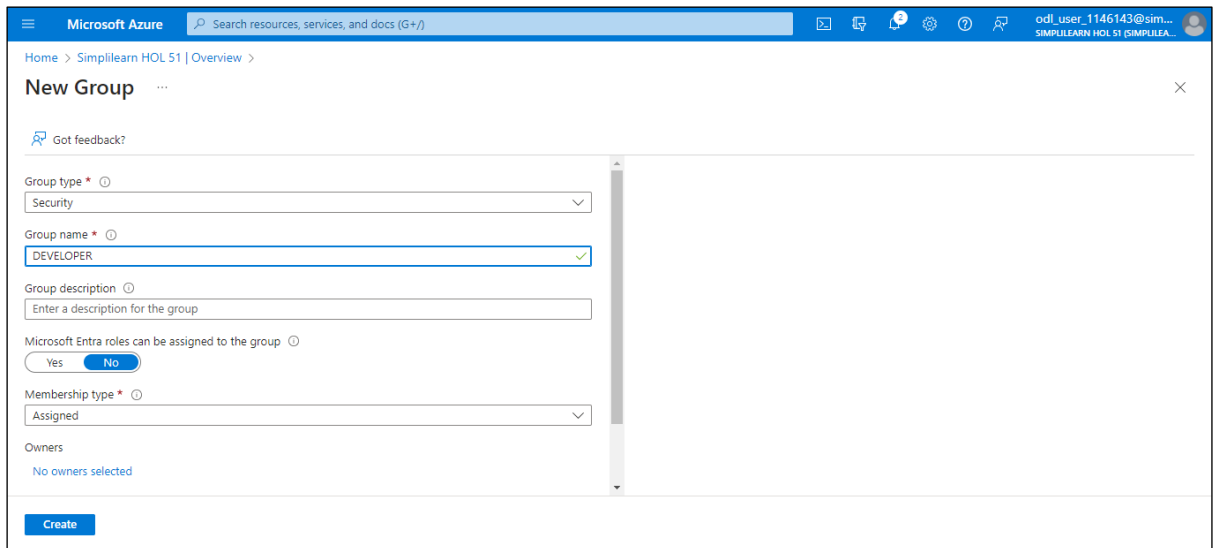
2.1 In the Home page, search for and select Microsoft Entra ID



2.2 In the **Overview** page, select **+Add** and then choose **Group**



2.3 In the **New Group** page, fill all the required details (Group name as **DEVELOPER**) as shown below and click on **Create**:



2.4 Similarly, create another group Group name as **SECURITY ANALYST** as shown below and click on **Create**:

Microsoft Azure

Home > Simplilearn HOL 51 | Overview >

New Group

Got feedback?

Group type *

Group name *

Group description

Microsoft Entra roles can be assigned to the group ☐ ☒

Membership type *

Owners
No owners selected

Create

The two groups, **DEVELOPER** and **SECURITY ANALYST** are now created.

Microsoft Azure

Home > Simplilearn HOL 51 | Groups >

Groups | All groups

Simplilearn HOL 51 - Microsoft Entra ID

New group Download groups Refresh Manage view Delete Got feedback?

Azure Active Directory is now Microsoft Entra ID. [Learn more](#)

Dynamic group memberships have not been updated due to system delays. We're working to resolve the issue.

Search Add filter

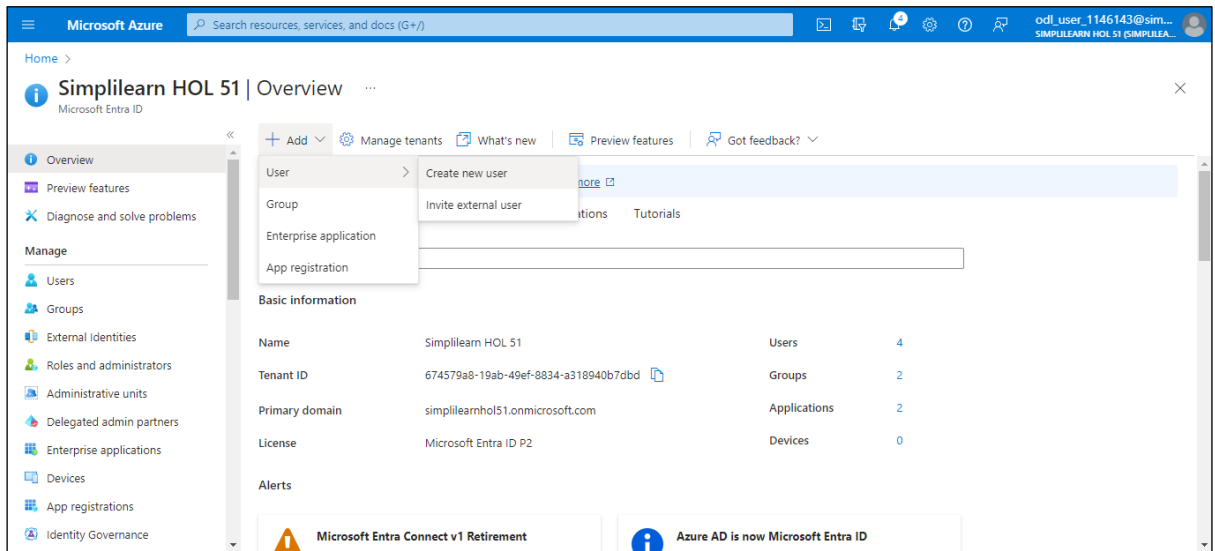
Search mode ☒ Contains

2 groups found

☐	Name ↑	Object Id	Group type	Membership type
☐	DE DEVELOPER	2ddc0923-a136-4932-ae59-5edfda57b852	Security	Assigned
☐	SA SECURITY ANALYST	087d0e95-1a02-4f8b-b1b6-1abd829ee03e	Security	Assigned

Step 3: Create two users and add them to each of the resource groups in Microsoft Entra ID

3.1 In the Overview page, click on **Add**, select **User** from the drop-down, and click on **Create new user**



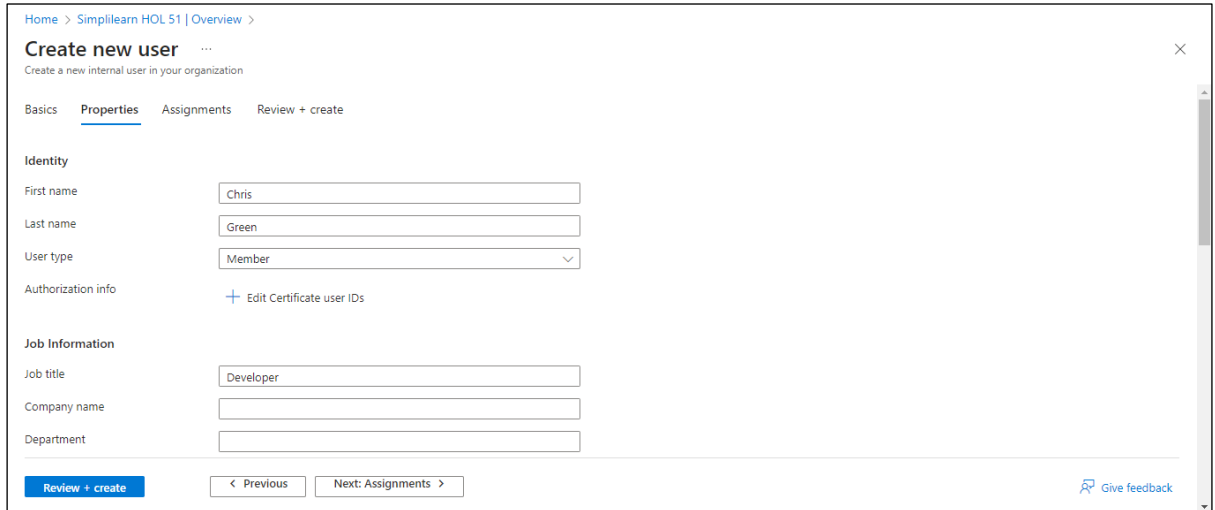
3.2 In the **Create new user** page, navigate to **Basics** tab and enter the below information:

The screenshot shows the 'Create new user' page in the Microsoft Azure portal. The 'Basics' tab is selected, and the form fields for creating a new user are visible. The form includes the following fields and options:

- User principal name:** A text box containing 'Chris' and a dropdown menu showing 'simplilearnhol51.onmicr...' with a 'Domain not listed' link.
- Mail nickname:** A text box containing 'Chris'.
- Display name:** A text box containing 'Chris Green'.
- Password:** A text box with a masked password '*****' and an 'Auto-generate password' checkbox.
- Account enabled:** A checkbox that is checked.

At the bottom of the form, there are buttons for 'Review + create', '< Previous', and 'Next: Properties >'. A 'Give feedback' link is also present in the bottom right corner.

3.3 Click on the **Properties** tab and enter **Job title** as **Developer**. Then, click on **Review + create**



Home > Simplilearn HOL 51 | Overview >

Create new user

Create a new internal user in your organization

Basics **Properties** Assignments Review + create

Identity

First name: Chris

Last name: Green

User type: Member

Authorization info: + Edit Certificate user IDs

Job Information

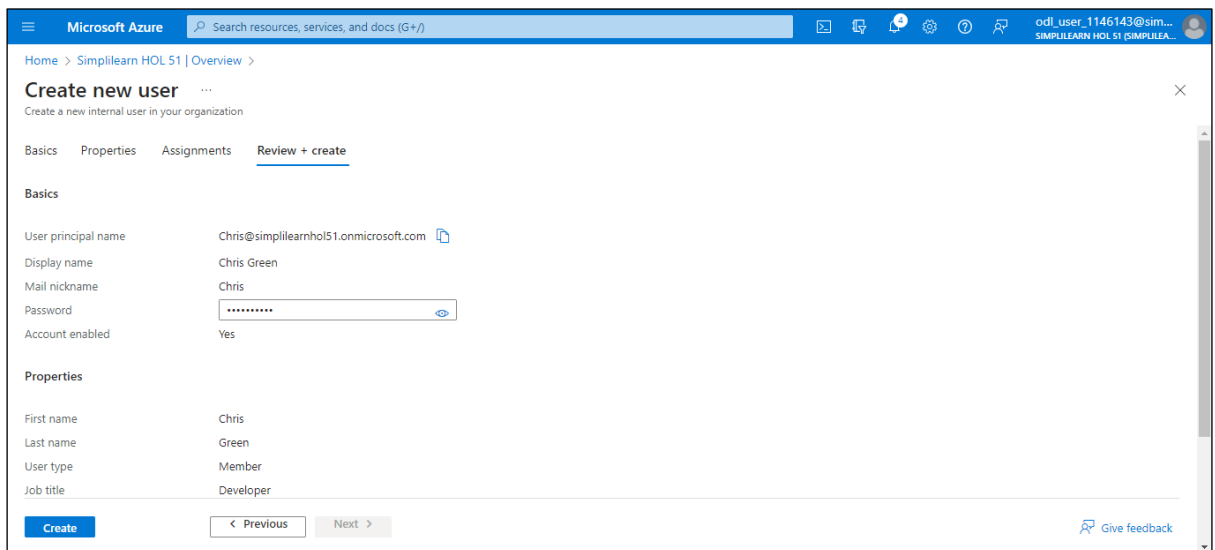
Job title: Developer

Company name:

Department:

Review + create < Previous Next: Assignments > Give feedback

3.4 In the **Review + create** tab, copy the auto-generated password provided in the **Password** box. This password is given to the user to sign in for the first time. Once the details are entered, click on **Create**.



Microsoft Azure Search resources, services, and docs (G+)

Home > Simplilearn HOL 51 | Overview >

Create new user

Create a new internal user in your organization

Basics Properties Assignments **Review + create**

Basics

User principal name: Chris@simplilearnhol51.onmicrosoft.com

Display name: Chris Green

Mail nickname: Chris

Password: [masked]

Account enabled: Yes

Properties

First name: Chris

Last name: Green

User type: Member

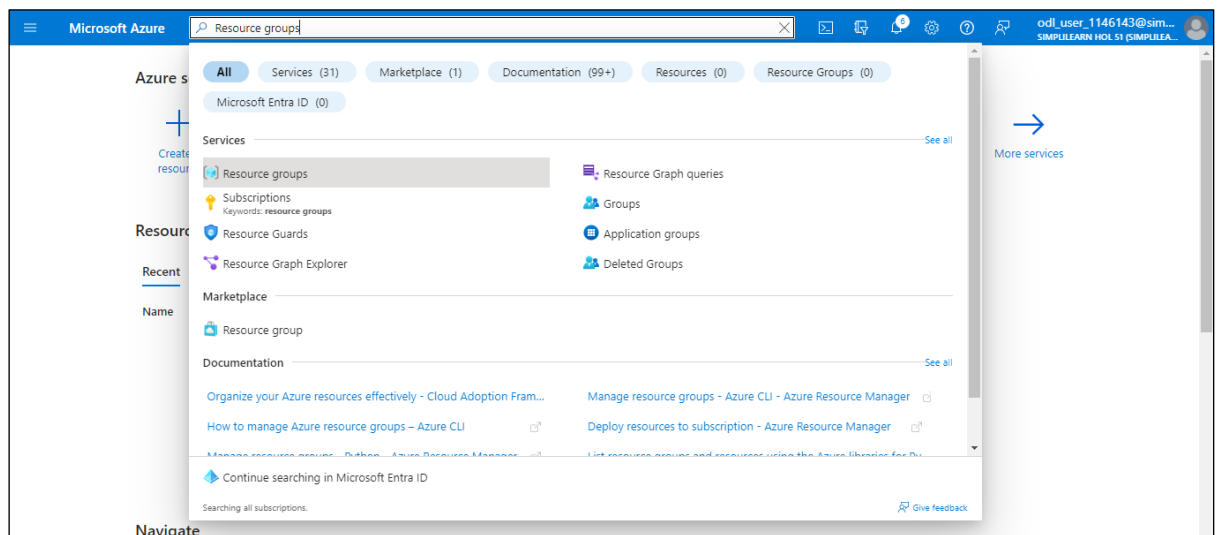
Job title: Developer

Create < Previous Next > Give feedback

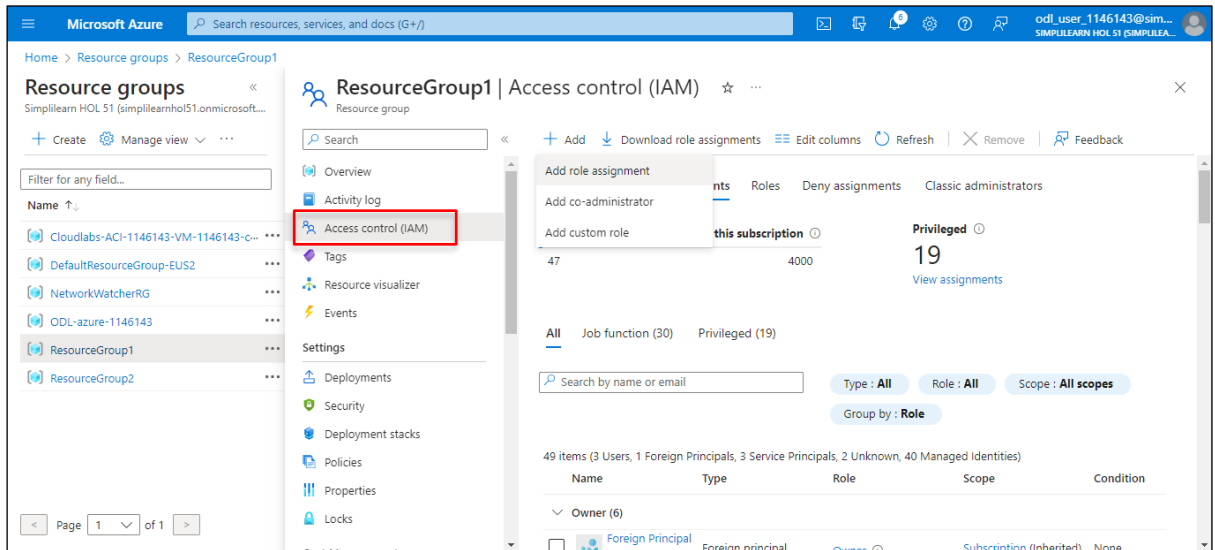
3.5 Similarly, create another user while assigning it to the **SECURITY ANALYST** group as shown below:

Step 4: Grant contributor and security assessment contributor access to the DEVELOPMENT and SECURITY ANALYST groups

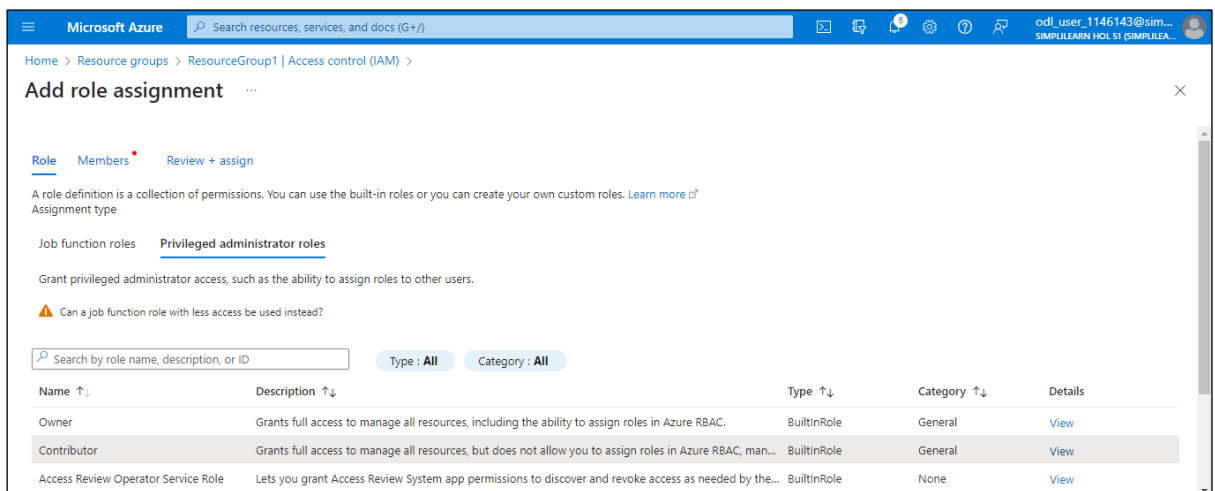
4.1 In the **Home** page, search for and select **Resource groups**



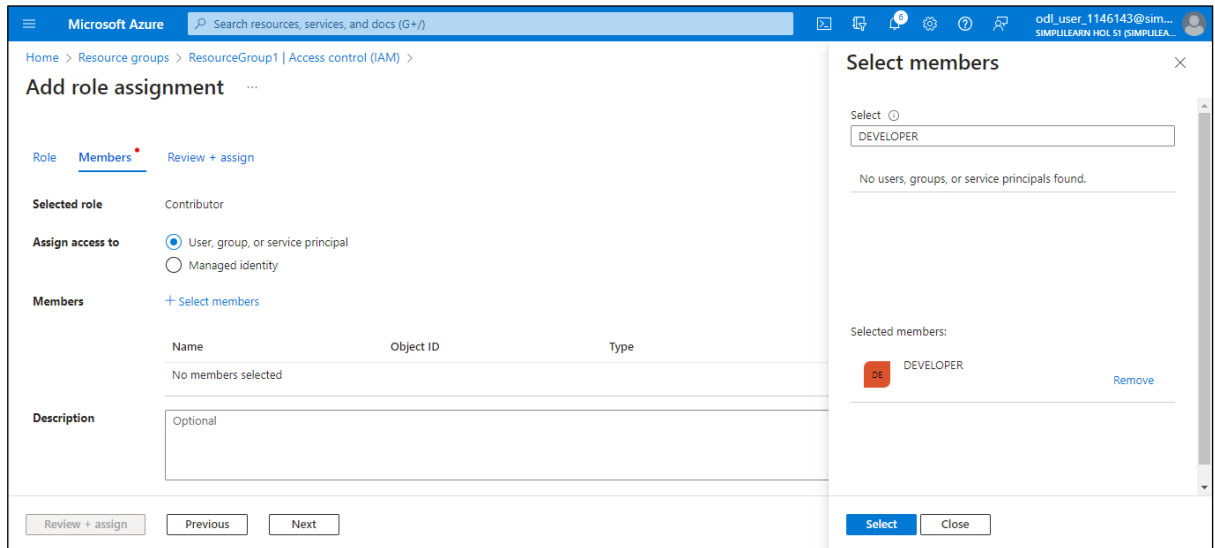
4.2 Click the specific resource group **ResourceGroup1** and click on **Access Control (IAM)**. Then click on **+Add** and select **Add role assignment**.



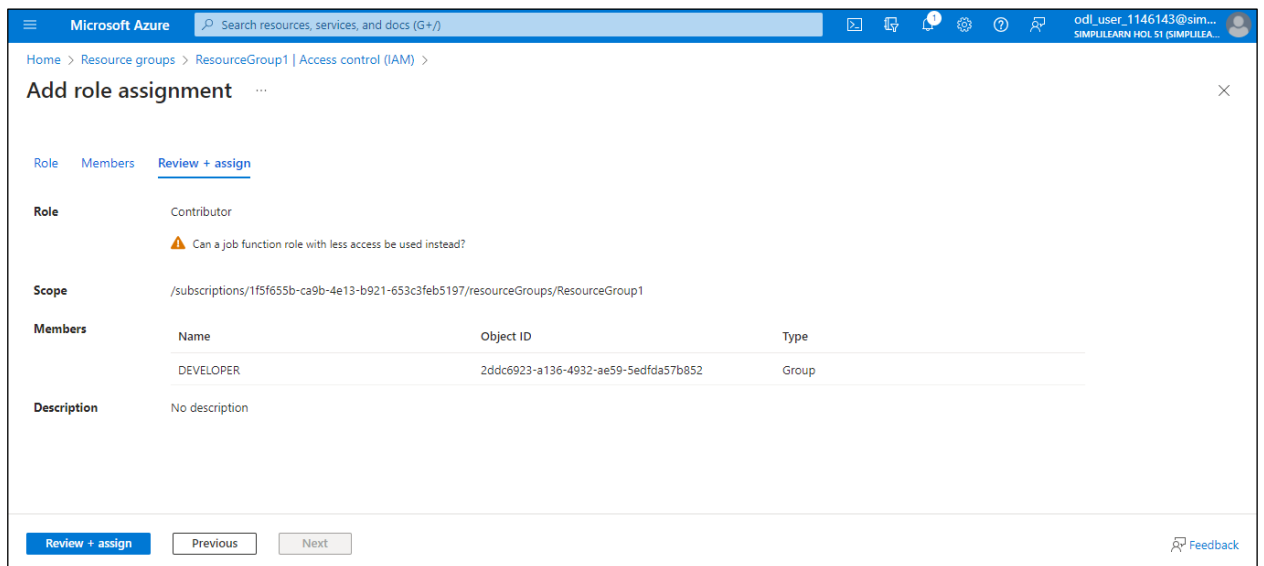
4.3 The **Add role assignment** pane opens. Click on the **Privileged administrator roles** and select **Contributor** from the list.



4.4 In the **Members** tab, click on **+Select members**. In the **Select members** pane, search for and select **DEVELOPER** and click on **Select**.



4.5 Select the **Review + assign** tab and click on **Review + assign**



4.6 Similarly, assign **Security Assessment Contributor** access to the **SECURITY ANALYST** group and click on **Select**

The screenshot shows the 'Add role assignment' page in the Microsoft Azure portal. The 'Members' tab is selected, and the 'Security Assessment Contributor' role is assigned to the 'SECURITY ANALYST' group. A 'Select members' sidebar is open, showing a list of members with 'SECURITY ANALYST' selected.

Name	Object ID	Type
No members selected		

4.7 Select the **Review + assign** tab and click **Review + assign**

The screenshot shows the 'Add role assignment' page in the Microsoft Azure portal. The 'Review + assign' tab is selected, and the 'Security Assessment Contributor' role is assigned to the 'SECURITY ANALYST' group. The 'Review + assign' button is highlighted.

Name	Object ID	Type
SECURITY ANALYST	087d0e95-1a02-4f8b-b1b6-1abd829ee03e	Group

By following the above steps, you have created two resource groups with two users and assigned appropriate roles to each group.